

Problem Set for 13 March 2025

Exercise 1 Let $f : U \rightarrow V$ be linear map and $\dim U = n$, $\dim V = m$.

1. Let $E \subset V$ be an k -dimensional subspace, prove that

$$\dim f^{-1}(E) \geq n - m + k.$$

2. When f is surjective, the equality holds.

Exercise 2 Read the following table and briefly talk about your understanding or confusion?

Concepts	Matrices	Linear maps
Linear transformation	$A_{m \times n}$	$\varphi : \mathbb{F}^m \rightarrow \mathbb{F}^n$
Basis	$(u_1, \dots, u_m)$, an invertible matrix	Omit.
Image of a basis	$(u_1, \dots, u_m) \cdot A$	$(\varphi(u_1), \dots, \varphi(u_m))$
	$E_{i,j}$	$\varphi(u_i) = v_j$, and $\varphi(u_{\neq i}) = 0$
Sum of maps	$A_1 + A_2$	$\varphi_1 + \varphi_2$
Scalar of maps	$\lambda \cdot A$	$\lambda \cdot \varphi$
Composition	$A \cdot B$	$\psi \circ \varphi$
V-composition	$\begin{pmatrix} A \\ B \end{pmatrix}$	$U \rightarrow V \times W$ $u \mapsto (\varphi(u), \psi(u))$
H-composition	$(A \ B)$	$U \times V \rightarrow W$ $(u, v) \mapsto \varphi(u) + \psi(v)$
D-composition	$\begin{pmatrix} A & O \\ O & B \end{pmatrix}$	$U_1 \times U_2 \rightarrow V_1 \times V_2$ $(u_1, u_2) \mapsto (\varphi(u_1), \psi(u_2))$
Dual map	A^T	$\text{Hom}_{\mathbb{F}}(V, \mathbb{F}) \rightarrow \text{Hom}_{\mathbb{F}}(U, \mathbb{F})$ $f \mapsto f \circ \varphi$
Inverse	A^{-1}	φ^{-1}
Adj	$\text{Adj}(A)$	$\bigwedge^{n-1} U \rightarrow \bigwedge^{n-1} U$ $\sum u_{i_1}^\lambda \wedge \dots \wedge u_{i_{n-1}}^\lambda \mapsto \sum \varphi(u_{i_1}^\lambda) \wedge \dots \wedge \varphi(u_{i_{n-1}}^\lambda)$
Rank	$r(A)$	$\dim \text{im } \varphi$
Equivalence	$A = P^{-1} \tilde{I}_r Q$	$\exists \{u_i\}_{i=1}^m, \{v_j\}_{j=1}^n : \varphi(u_{i \leq r}) = v_i$, otherwise 0.

Concepts	Matrices	Linear maps
Injection	Full row rank	Omit.
	$\exists B, AB = I$	Omit
Surjection	Full column rank	Omit.
	$\exists B, BA = I$	Omit.
Fundamental decomposition	$A = R \cdot C$, where R has full row rank, C has full column rank	φ is always the composition of a surjection (from its domain) and an injection (to its codomain).
Elementary transformation	$S_{i,j}$	φ swaps u_i and u_j
	$I + \lambda E_{i,j}$	$\varphi(u_j) = v_j$ for $j \neq i$, and $\varphi(u_i) = u_i + \lambda u_j$.
	$I + (\lambda - 1)E_{i,i}$	$\varphi(u_j) = v_j$ for $j \neq i$, and $\varphi(u_i) = \lambda v_i$.
Projection	Take the first r columns	$\pi \circ \varphi$, where π is the projection $\text{span}(\{v_i\}_{i=1}^n) \rightarrow \text{span}(\{v_i\}_{i=1}^r)$.
Inclusion	Add k zero columns	$\iota \circ \varphi$, where ι is the inclusion $\text{span}(\{v_i\}_{i=1}^n) \rightarrow \text{span}(\{v_i\}_{i=1}^{n+k})$.
Nilpotent	$A^N = O$	Omit
Similarity	$A = P^{-1}BP$	$\exists$ base change $(u_i) = (v_i) \cdot P$, such that the linear map has matrix form B .
Invariant subspace	$A \sim \begin{pmatrix} A_1^{(r \times r)} & A_2 \\ O & A_4 \end{pmatrix}$	$\exists U' \subset U : \varphi(U') \subset U'$.
Quotient an invariant subspace	$\begin{pmatrix} A_1^{(r \times r)} & A_2 \\ O & A_4 \end{pmatrix} \mapsto A_4$	$\varphi(x + U) = \varphi(x) + U$.
Mutual annihilation	$AB = O_{m \times m}$, and $BA = O_{n \times n}$	See example class last year.
Blocks	$A \sim \begin{pmatrix} A_1 & O \\ O & A_2 \end{pmatrix}$	$U_1 \oplus U_2 = U$, such that $\varphi(U_i) \subset U_i$.
Upper triangulation	$A \sim T$	There exists a flag of U_i with $U_i \subset U_{i+1}$, $\dim U_i = i$, such that $\varphi(U_i) \subset U_i$.
Diagonalisation	$A \sim P^{-1}\Lambda P$	Omit.
Fitting lemma	$A \sim \begin{pmatrix} N & O \\ O & M \end{pmatrix}$, where N is nilpotent, and M is invertible.	For $\varphi : U \rightarrow U$, $\text{im}(\varphi^N)$ and $\ker(\varphi^N)$ are invariant subspaces for large N . In particular, $U = \text{im}(\varphi^N) \oplus \ker(\varphi^N)$.

Concepts	Matrices	Linear maps
Tensor	Kronecker product	Tensor product

Exercise 3 (optional) Let $f, g : \mathbb{C}^{2024} \rightarrow \mathbb{C}^{2025}$ be linear injections, such that $af + bg$ is a surjection for any non-zero a and b .

1. One can find u, v such that $f(u) \notin \text{im}(g)$ and $g(v) \notin \text{im}(f)$. Show that $\{u, v\}$ is a linearly independent set.
2. Find bases such that the matrix forms of f and g are

$$f \mapsto \begin{pmatrix} I_{2024} \\ \mathbf{0}^T \end{pmatrix}, \quad g \mapsto \begin{pmatrix} \mathbf{0}^T \\ I_{2024} \end{pmatrix}.$$

3. Say $(\mathbb{C}^{2024} \xrightarrow{\alpha} \mathbb{C}^{2024}; \mathbb{C}^{2024} \xrightarrow{\beta} \mathbb{C}^{2025})$ is a good pair of linear endomorphisms, provided

$$\beta \circ (af + bg) = (af + bg) \circ \alpha, \quad (\forall a, b \in \mathbb{C}).$$

Prove that $(\alpha; \beta)$ forms a linear space, and find its dimension.